



3D numerical simulation of tsunami generation and propagation, case study: Makran tsunami generation and penetrating in Chabahar Bay

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ABSTRACT

The Makran subduction located at the northwest of the Indian ocean, at the Oman Sea mouth has been and is the source of massive tsunamis that threatens the coasts of Iran, Pakistan, India and Oman. The initial tsunami wave produced by the magnitude and focal depth of Makran submarine earthquake in 1945 is determined based on the Okada algorithm. 3D propagation of the generated mega wave in the Indian Ocean and Oman Sea is simulated by adopting OpenFOAM open source CFD software, in its global sense and verified vs. the available data. The wave characteristics are extracted from the global model at the mouth of Chabahar Bay are inserted as the input in a regional model which consist of precise geometry and bathymetry of the bay and finer mesh size. The wave run-up is predicted at the main ports at the Indian Ocean and Oman Sea coastline. The obtained results are implemented to assess the tsunami risk zone of Teess beach town, at Chabahar Bay coastline. The advanced procedure applied in this research consists of the accurate production of the original tsunami and three stages of global and regional 3D and local 2D propagation simulation, that provides reliable results in more accurate estimation of wave run up height and tsunami risk zone, for practical engineering applications.

1. Introduction

Tsunamis mainly occur due to earthquakes at subduction zones and induce gigantic disturbances on ocean surface (Heidarzadeh et al., 2011). After the tragic tsunamis of Indian Ocean near the coast of Sumatra island, Indonesia in 2004 and Pacific coast of Tohoku, Japan in 2011, which took many lives (about 250,000 dead) and caused great economic damage, many studies are run to advance knowledge of the experts of this field of science on the physics of the phenomena and improve the models to simulate the generation and propagation of the tsunami waves towards and the run-up therein over coastlines (Okal et al., 2008; Grilli et al., 2013). Consequently, it is crucial to improve the efficiency of the associated numerical models applied for predicting and estimating the tsunami run-up and the resulting flooded zone.

Makran subduction, located at Arabian Sea, the Northwest of Indian ocean at the mouth of Oman Sea and adjacent to the coasts of Iran, Oman and Pakistan, is possibly the second source of massive tsunamis' occurrence in Indian Ocean (Heidarzadeh et al., 2009). The fault line is shaped by the impact of the Arab and Eurasian tectonic plates and extends approximately 900 km from the Strait of Hormuz to Karachi port in Pakistan. The location of the main earthquakes originated from the

Makran subduction zone (MSZ) and the year of the occurrence are illustrated in Fig. 1. The major tsunami due to MSZ occurred in 1945, which left at least 4000 dead from Pakistan, India, Iran and Oman. The coordinate of subduction is 63.601° longitude and 24.927° latitude. The tsunami occurred at 15 km focal depth and 8.1 Richter magnitude (Ambraseys et al., 1982).

The threat of the tsunami due to MSZ to the Iranian coast in the Northern Oman Sea is quite perceptible, particularly at the coasts of Chabahar Bay, with high population concentration. Next to, the port of Chabahar, located at the mouth of the bay has a strategic contribution in the regional economy development. The location of the Chabahar Bay and the recorded major earthquakes in the region, with magnitudes greater than 4 Richter, are shown in Fig. 2.

The generation, propagation and predicting the run-up, estimating the hazard zone and application of preventing measures of tsunami has been and is the theme of the recent studies. This subject requires multi-disciplinary approaches, where seismologist and geologist to oceanographers and coastal engineers are involved. In terms of tsunami wave generation, by assuming a number of simplifying measures like a rectangular shape for the fault, Mansinha and Smylie (1971) presented a numerical solution for the governing equations of the of the elastic

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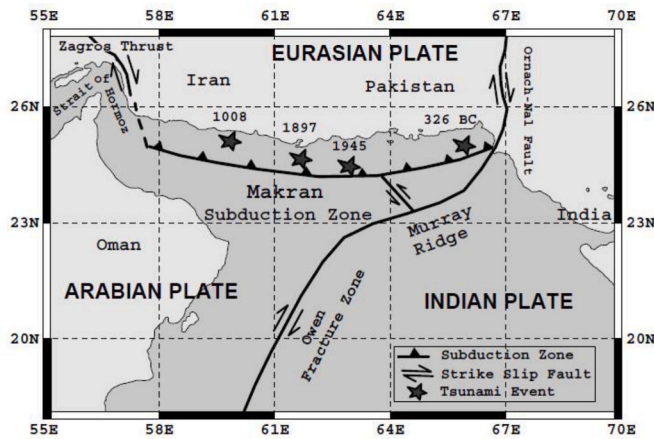


Fig. 1. The historical record of earthquake and the year of occurrence due to Makran subduction zone (MSZ) (Heidarzadeh et al., 2011).

homogeneous space deformation and sea bed shape deformation due to earthquake. One of the most significant attempts made in predicting tsunami generation is made by Okada (1985), where he presented an algorithm for calculating the 3D sea bed deformation due to earthquake with a specified magnitude and focal depth. Later in 1992, he presented an algorithm for calculating the internal displacements, strains and tilts by solving the governing equations of elastic homogeneous space (Okada, 1992). In his introduced algorithm, the effect of equilibrium equations for motion of bed sheets on each other due to earthquake in the focal depth is of major concern and the displacement rate and motion angle of the plates on each other are specified. Wells and Copper-smith (1994) collecting field data of many subductions and fitting the best mathematical relationship, proposed equations to calculate the fault length, width and displacement in terms of earthquake magnitude.

In terms of tsunami wave propagation, solitary waves with infinite wavelength, introduced by John Scott Russell (1834), is usually substituted with the tsunami wave. There exist many numerical models, calibrated and validated through experimental studies to simulate the solitary wave propagation and estimate run-up (Synolakis, 1987; Hsiao et al., 2010; Zandi et al., 2017; Rafizadeh et al., 2019). Later the N-wave was introduced as the better replication for tsunami. Tadeipalli et al. (1994) compared the solitary wave run up and corresponding N-wave, in an experimental and numerical manner, where the maximum run-up of some isosceles N-wave and solitary wave with the same amplitude are measured and the results reveal that the maximum isosceles N-wave run-up is 1.4 times greater than solitary wave run-up with the same amplitude.

Recently, many researches simulated real tsunamis. Heidarzadeh et al. (2007) assessed the initial profile of the tsunami induced by 1945 Makran subduction (MSZ) and the wave propagation in the Indian

Ocean. They simulated the tsunami propagation by adopting the TUNAMI s/w, which solves the continuity and momentum equation by applying the shallow water approximations. The results indicate that the tsunami generation risk from Makran subduction zone can be of three main categories: very little for earthquake with magnitudes up to 7, little to medium for earthquake with magnitudes within 7 and 7.5, and high for magnitudes greater than 7.5. Okal and Synolakis (2008) defined two worst-case scenarios for the possible mega-thrust earthquakes in the MSZ and assessed their far-field effects across the Indian Ocean basin. They simulated 3D propagation of tsunami wave in Indian Ocean for a number of scenarios of mega-thrust earthquakes, within probable to possible range. Ha et al. (2009) simulated the tsunami propagation in the South China Sea which potentially originated from a giant rupture along the Manila Trench. Tsunami run-up height and arrival time of the major coasts are predicted through the TUNAMI-N2-NUS model. This model is improved version of the original TUNAMI model where more phenomena like astronomic tide, sea bottom friction, dispersion, Coriolis force, and spherical curvature are of concern. The results indicate that the Sunda shelf and the Natuna islands may act as natural barriers, sheltering southwest part of the south China sea and Singapore strait. Grilli et al. (2013) simulated generation and propagation of the near-field and far-field impact of the 2011 Tohoku tsunami by applying the Boussinesq-type model. The results indicate that dispersive effects are negligible in the near-field, due to the short propagation distances and coarse grid resolution, while may account for 20–40% of tsunami

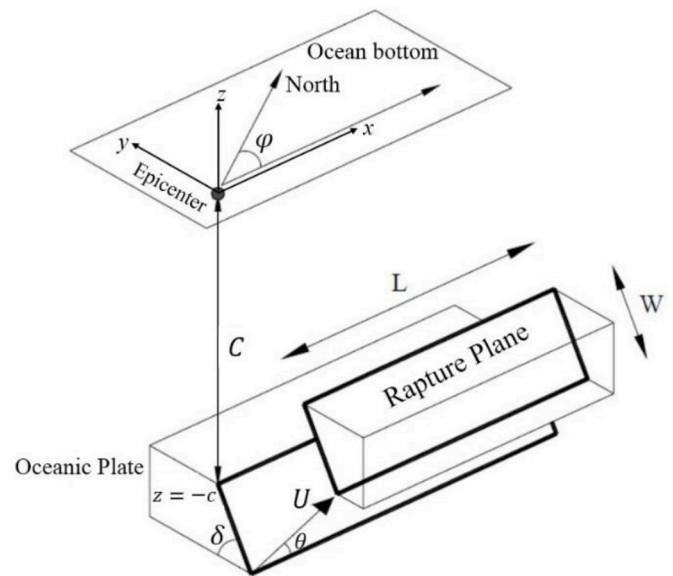


Fig. 3. Fault motion parameters.

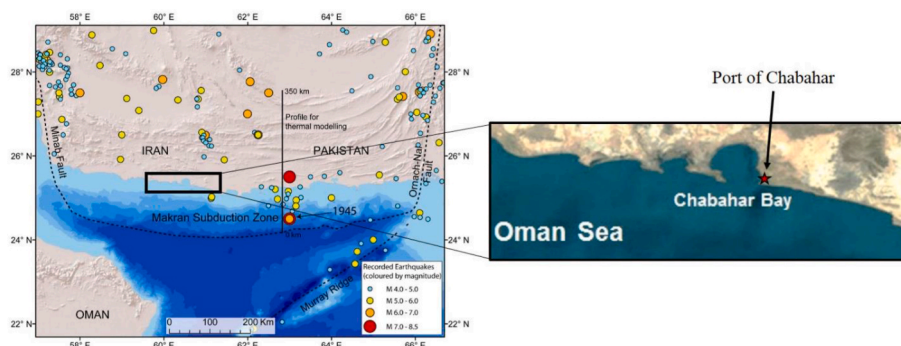
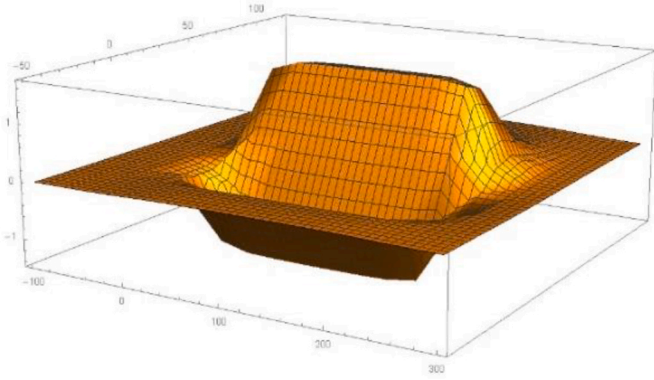


Fig. 2. Location of Chabahar bay at Northern Oman Sea coastline.

Table 1

Comparison of the field data and the estimations obtained through the Okada algorithm for some real subduction events.

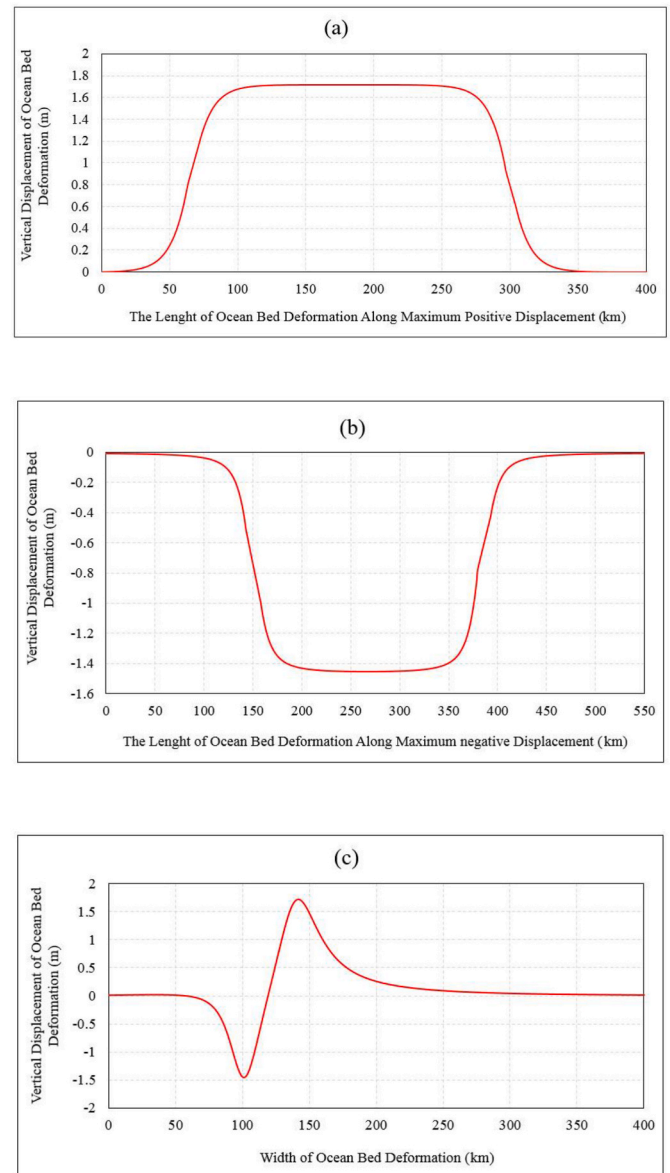
Event Name	Quake Magnitude	Rupture (km)		Angles (Deg)			Depth (km)	Displacement (m)	Uplift (m)		Error (%)
		Length	Width	Dip	Strike	Slip			Actual	Model	
Portorico (1918)	7.3	66	23	70	205	106	4	4	+0.8	+0.74	+7.5
Mexico (1995)	8.0	160	60	16	309	90	10	4	+1.85	+1.79	+3.24
Mexico (1962)	7.1	40	35	25	296	90	12	0.65	+0.22	+0.2	+9
Indonesia (2004)	9.3	443	170	8	329	110	25	30	+10.0	+9.32	+6.8

**Fig. 4.** 3D seabed vertical deformation of 1945 Makran subduction ($M = 8.1$ Richter and $d = 15$ km, and $\delta = 7^\circ$).

amplitude in deep water, thus the Boussinesq model application is justified. Rasyif et al. (2019) by applying COMCOT-SED model, simulated the tsunami wave propagation around Banda Aceh (Sumatra island, Indonesia), where by coupling a hydrodynamic model with a sediment transport module, the coastal morphological changes are assessed. The Cornell Multigrid Coupled Tsunami Model (COMCOT), which is based on the non-linear shallow water equations (NSWE) was coupled with the XBeach Model to simulate the hydrodynamic processes and sediment transport in a simultaneously manner. The coupled model is known as COMCOT-SED. The results of this study indicate that the results are consistent with the field data where the tsunami waves caused offshore erosion.

It is revealed that in these studies, the physical phenomena are simplified either in the simulation of generated wave or in the applied assumptions for propagation. The NSWEs which is the base of the conventional tsunami models are derived from the depth-integrating Navier–Stokes equations, provided that the horizontal length scale is much greater than the vertical length scale. Under this circumstance, mass conservation implies that the vertical velocity scale of the fluid is small compared to the horizontal velocity scale. through the momentum equation It can be shown that the vertical pressure gradients are hydrostatic. NSWE is a good model for the propagation of tsunami throughout the open ocean, away from shore. When a vertical velocity term is not present in the shallow water equations, this velocity is not necessarily zero close to the shore, where the wave compresses horizontally and grows vertically, and when waves experience abrupt depth change. The 3D approach provides the advantage of taking into account the vertical velocity and the alteration of pressure from hydrostatic, particularly for studies run on assessing the potential influences of introduced coastal barriers against tsunami wave run-up.

The objective of this article is to adopt and couple the state of the art 3D models of tsunami wave generation and propagation. For this purpose, the MSZ is considered as the case study. The initial tsunami wave is produced based on the Okada algorithm, in a 3D form. The propagation of tsunami is simulated by the computational fluid dynamics package, Open Field Operation and Manipulation (OpenFOAM). The performance of OpenFOAM s/w in simulating free surface flow and the 3D

**Fig. 5.** The 2D presentation of vertical sea bed deformation along the length (a and b) and orthogonal to the subduction zone (c) of 1945 Makran tsunami ($M = 8.1$ Richter, $d = 15$ km, $\delta = 7^\circ$).

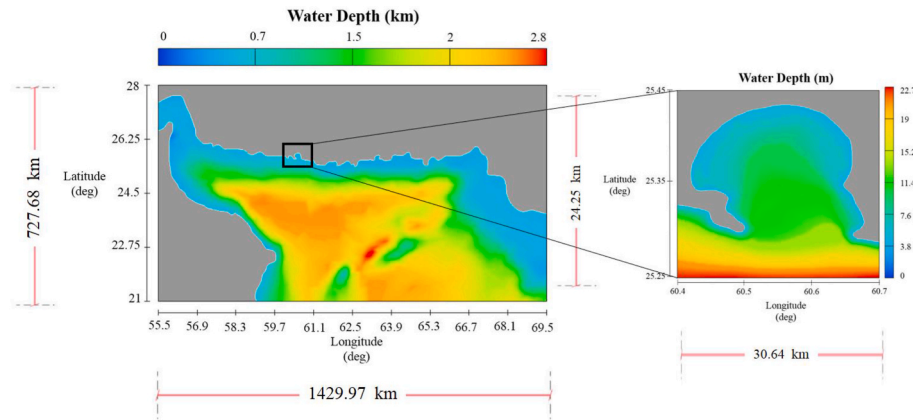


Fig. 6. Bathymetry and model domain, (a) global model; Indian Ocean, (b) regional model; Chabahar Bay.

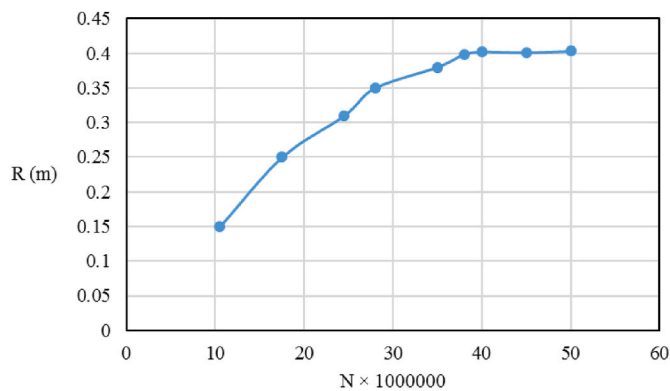


Fig. 7. Mesh independence study for numerical simulation of the global model, where R is the run-up at Chabahar Port.

propagation of a tsunami wave, are already assessed and proved through comparison with extensive experimental data provided by [Higuera et al., \(2013\)](#); [Higuera et al., \(2014\)](#); [Honarmand et al., \(2018\)](#). In this study, the precision of simulating the wave transmission is improved stage by stage, begin from the global model, Indian Ocean to the regional model, Chabahar Bay, town of Teess, and is applied in the engineering practices.

This article is structured as follows: 3D numerical simulation of tsunami wave generation and propagation are described in Secs. 2 and 3, respectively. The results and discussion are presented in Sec. 4 and the article findings are concluded in Sec. 5.

2. 3D numerical simulation of tsunami wave generation

2.1. Okada algorithm

The production and simulation of the wave generation are considered as the first attempts in the studies run on characteristics of a tsunami. The most important cause of tsunami is earthquake at the ocean bed due to bed subduction and motion of the tectonic plates on each other. Because water is incompressible, the deformation of the bed is transferred to the surface forming the tsunami wave which spreads in all directions from the generating zone. The direction of the main energy propagation is generally orthogonal to the direction of the fault.

By applying Okada's algorithm, the deformation of each point at the ocean bed can be calculated in its 3D form. The fault geometry and applied parameters in Okada algorithm are shown in [Fig. 3](#), where, δ is the fault landslide angle, θ is the angle between the fault displacement vector and the length of the fault failure plate and ϕ is the angle between

the north and the fault direction.

Based on the Okada algorithm, by specifying the earthquake magnitude (M) in Richter scale and the focal depth (C) in kilometer, length (L), width (W), fault deformation (U), and subsequently the vertical displacement of each point ($u_z(x, y, z)$) on ocean bed is calculated (Wells and Coppersmith, 1994).

The results of Okada algorithm are verified through comparison with field data for a number of real ocean subductions. The events applied in this study for the verification include Portorico 1918 tsunami assessed by [Mercado and McCann \(1998\)](#); Mexico 1962 tsunami reported by [Ortiz et al. \(2000\)](#); Mexico 1995 tsunami assessed by [Ortiz et al. \(2000\)](#); and the mega-tsunami of 2004 in Indonesia assessed by [Yalciner et al. \(2005\)](#). The results are tabulated in [Table 1](#), where the Okada algorithm is promising method in estimating ocean bed uplift deformation, with errors less than 9%.

2.2. Modeling the initial wave shape of the 1945 Makran tsunami

To model the Makran 1945 deformed seabed through Okada algorithm, the seismic parameters therein estimated by [Byrne et al. \(1992\)](#), as being: the ocean bed is of 7° dip angle (δ) and 90° strike angle (θ) between the fault displacement vector and the failure plane length are applied. The earthquake magnitude is 8.1 Richter and the focal depth is 15 km. By applying Okada algorithm in Mathematica s/w the sea bed deformation is simulated, which is considered as the initial tsunami wave profile. The 3D seabed vertical deformation of 1945 Makran subduction, extracted from the model is presented in [Fig. 4](#).

[Page et al. \(1979\)](#) found that the maximum sea floor deformation of Makran (1945) subduction was estimated at 2 m, while according to this simulation, the maximum positive vertical displacement is estimated at 1.714 m, which is in reasonable agreement with the existing reports about the sea bed uplift of the 1945 event at 14.3% error. The maximum negative vertical displacement is estimated at 1.453 m. The maximum and minimum bed vertical deformation due to Makran subduction along the fault line (parallel) and in orthogonal direction (i.e. the direction of the propagation), are shown in [Fig. 5](#).

In this study, in simulating tsunami propagation, only the 3D ocean bed deformation rate in the vertical direction (z) is of concern and bed movement in the horizontal directions (x and y) are not.

3. 3D numerical simulation of tsunami wave propagation

3.1. OpenFOAM s/w and model set up

OpenFOAM is a C++ toolbox to develop customized numerical solvers, and pre or post processing utilities for the solution of continuum mechanics problems, including computational fluid dynamics (CFD). Some advantages of this software are: zero license costs, friendly

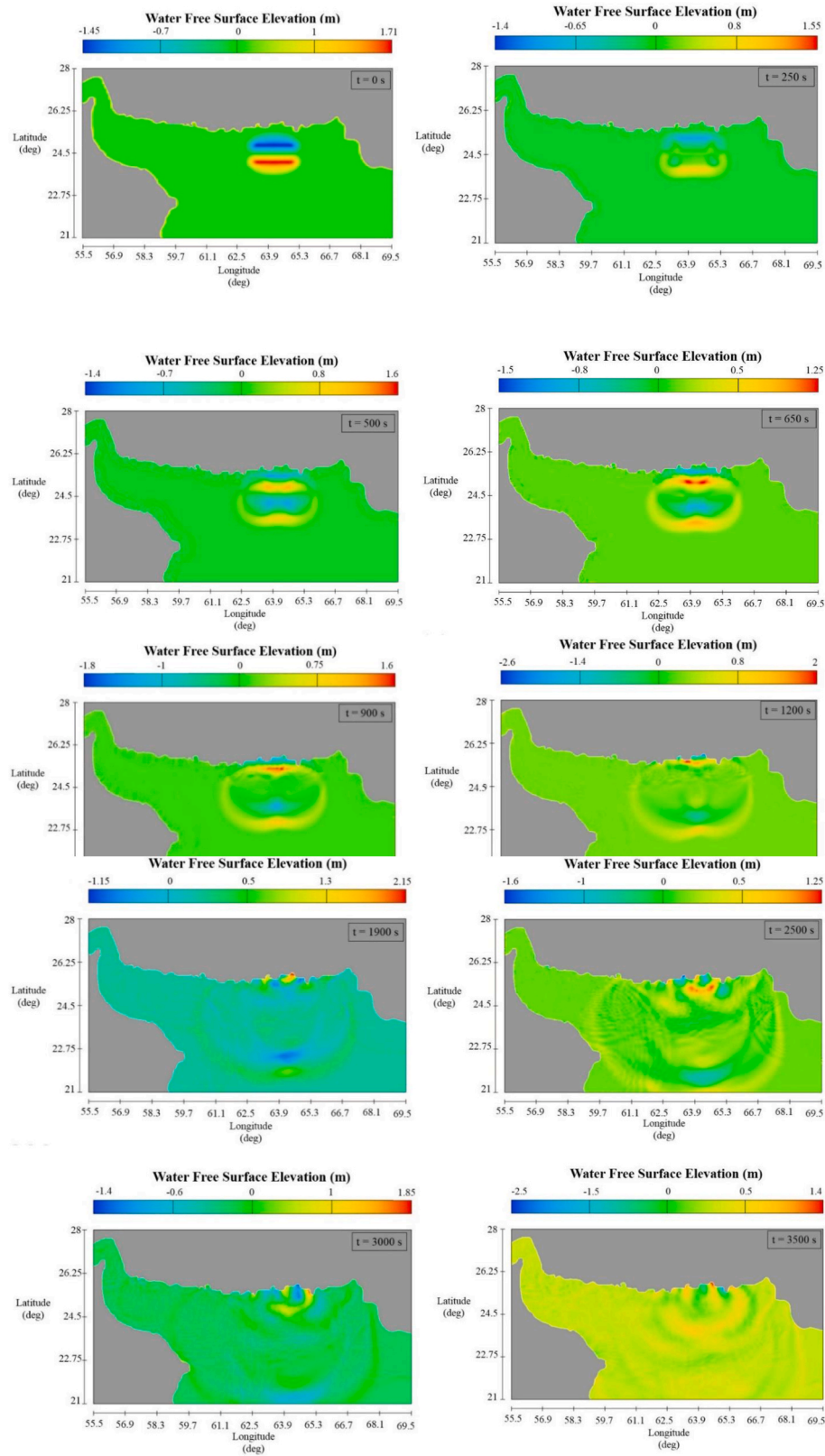


Fig. 8. Steps in tsunami wave propagation in the Indian Ocean, results extracted from the OpenFOAM global model simulation.

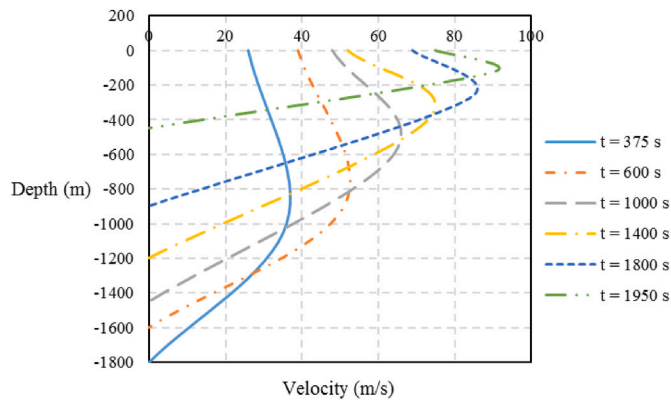


Fig. 9. Velocity profiles of tsunami propagating toward shoreline.

organization for partial differential equations, unstructured polyhedral grid capabilities and code customization according to problems. OpenFOAM solvers include incompressible flow with RANS and LES capabilities, compressible flow solvers with RANS and LES capabilities, buoyancy-driven flow solvers, DNS and LES, multiphase flow solvers, particle-tracking solvers, solvers for combustion problems, solvers for conjugate heat transfer, solid dynamic solvers. The OpenFOAM utilities are subdivided into mesh conversion, parallel processing utilities and post-processing utilities (OpenFOAM Programmer's Guide, 2017).

The fundamental governing equations in OpenFOAM open source software consist of the continuity and nonlinear momentum equations, as follow:

$$\nabla \cdot \vec{U} = 0 \quad (1)$$

$$\rho \frac{\partial \vec{U}}{\partial t} + \nabla \cdot (\vec{U} \vec{U}) - \nabla \cdot (\nu \nabla \vec{U}) = -\frac{1}{\rho} \nabla p + \vec{g} \quad (2)$$

where, $\vec{U} = \langle u \ v \ w \rangle^T$ is the velocity field, p is the piezometric pressure distribution function and ν is the fluid kinematic viscosity, ρ is the density and $\vec{g}$ is the gravity vector. The algorithm applied in OpenFOAM for discretization of the governing equations is the Pressure Implicit with Splitting of Operator (PISO) algorithm. In this study, the OpenFOAM 2.1.1 installed on the Linux operating system and the SUSE Studio GeekoCFD environment are applied. The test channel, a rectangular cube, reproduced by the blockMesh command and the shape of the coast slope are produced by the snappyHexMeshDict tool.

The IHFOAM solver is applied in simulating the free surface flow. The k- ϵ model is applied in the modeling turbulence. The k- ϵ turbulence model is the most common model applied in Computational Fluid

Dynamics (CFD) to simulate mean flow characteristic features for turbulent flow conditions. The k- ϵ model is a two equation model that gives a general description of turbulence by means of two transport equations. The original impetus for the k- ϵ model are to improve the mixing-length model, and to find an alternative to algebraically prescribing turbulent length scales in moderate to high complexity flows. (OpenFOAM Programmer's Guide, 2017; Islam and Soares, 2018; Higuera et al., 2014). The two folders of k and epsilon are added to the original subfolder 0. In the k folders, the wall function $kqRWallFunction$ is applied for both the wall surface and function and the $turbulentIntensityKineticEnergyInle$ is applied for the input boundary condition. In the epsilon folder, the wall function, $epsilonWallFunction$, is applied for wall surface and the wall function $turbulentMixingLengthDissipationRateInlet$ is applied for the input surfaces. The wall function $inletOutlet$ is applied for the output boundary condition in both files. Moreover, to add the surface roughness to the coast slope, the nut folder is added to the original subfolder 0. In this folder, the $nutkRoughWallFunction$ is applied for the surfaces which are intended to add roughness.

The boundary conditions which govern the simulations are consistent with patch inlet boundary condition for wave generation, atmosphere condition for the boundary of the free surface of the fluid and the wall boundary condition for other surfaces. Hence, the time dependent variations in the input velocity of flow are introduced into the software through the groovyBC library. The ParaView open source software is applied to visualize the results.

OpenFOAM is considered as a reliable and efficient tool for numerical solution of Navier-Stokes equations in complex boundary conditions including free surface unsteady flows and wave shoaling and

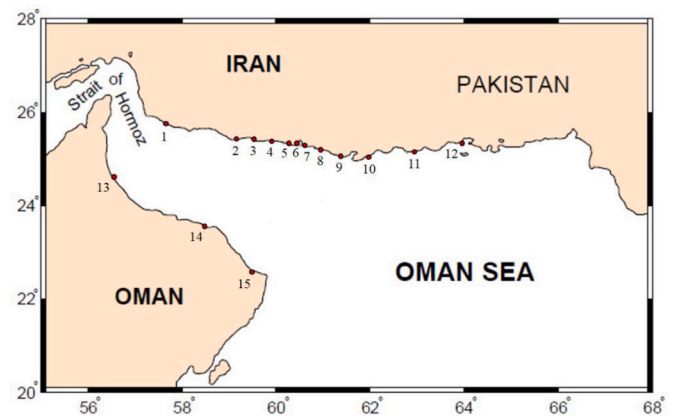


Fig. 10. Locations of major ports at the Indian Ocean and the Oman Sea affected by MSZ.

Table 2

The maximum run-up height and arrival time of 1945 Makran tsunami at a number of important ports.

Port Number	Country	Port Name	Longitude (deg)	Latitude (deg)	Maximum Wave Run-up (m)	Wave Arrival Time (s)
1	IRAN	Jask	57.76	25.62	0.012	2400
2		Kereti	59.09	25.34	0.035	2350
3		Kalak	59.37	25.29	0.055	2300
4		Bir	59.81	25.32	0.082	2250
5		Gurdim	60.12	25.29	0.135	2200
6		Konarak	60.42	25.3	0.11	2550
7		Chabahar	60.6	25.22	0.398	2100
8		Beris	61.17	25.05	0.75	1800
9		Jwani	61.75	24.92	1.55	1500
10	PAKISTAN	Gwadar	62.33	25.02	1.78	1200
11		Pasni	63.49	25.06	3.95	1100
12		Ormara	64.64	25.08	4.25	1000
13	OMAN	Suham	56.83	24.36	0.033	2150
14		Mascot	58.65	23.64	0.047	1950
15		Sur	59.54	22.64	0.055	1800

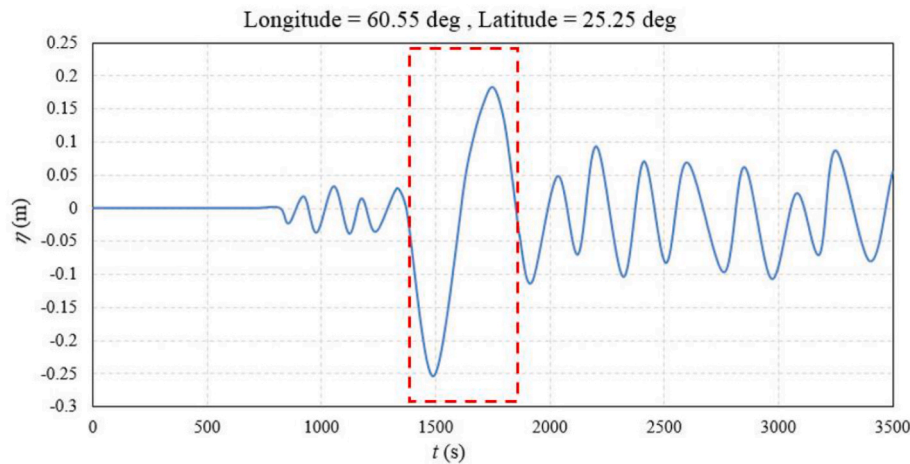


Fig. 11. Tsunami wave train at the mouth of Chabahar Bay extracted from the global model.

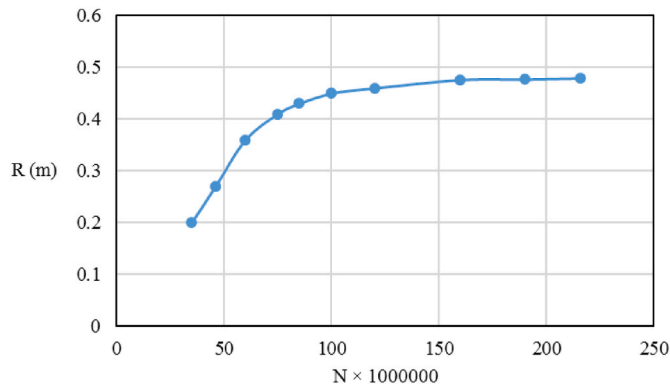


Fig. 12. Mesh independence study for numerical simulation of the local model, where R is the run-up at Chabahar Port.

attenuation (Higuera et al. 2013, 2014; Thaker et al., 2013; Islam et al., 2018; Adjali et al., 2015; Hu et al., 2016). The extensive comparative study of Honarmand (2017) reveals that water surface profiles of solitary waves (the long wave which is analogous in behavior to tsunami) are simulated by OpenFOAM in their realistic sense and are in agreement with data and as to run up, the predictions are found appropriate with a high level of accuracy; that is less than 5% discrepancy between the model and experiments for non-breaking and breaking wave propagation over sloped beaches.

3.2. Numerical simulation of 1945 Makran tsunami propagation

The tsunami propagation is numerically simulated in two stages: the spread of 3D generated wave is modeled in Indian Ocean and Oman Sea through a global model consisting of large resolution mesh network. The water surface undulation is extracted from the global model at the mouth of Chabahar Bay and introduced as the input to the regional model of Chabahar Bay with a relatively finer mesh size and accurate seabed bathymetry. In the global model, the ocean bed bathymetry data is obtained from the General Bathymetry Chart of the Oceans (GEBCO) digital atlas (IOC, IHO, BODC. 2003). For the regional model the data from the local high resolution (25 m) bathymetric surveying is applied. The bathymetry of the global and regional models, and the domains of the model set up are shown in Fig. 6.

4. Results and discussion

4.1. Global model

The initial tsunami wave produced by Okada algorithm is inserted into the model by applying the *SurfaceToCell* code in the OpenFOAM s/w. Mesh independence studies are run to derive an optimal mesh resolution, with considering convergence of run-up as the criterion for the acceptable level of sufficiency. The result is shown in Fig. 7.

The optimal size of the meshes in the global model in x and y directions is 1.4×1.4 km. In OpenFOAM s/w to make the meshes finer, sheet plates are applied. In BlockMesh folder the elevation of sheet plates is defined and the mesh size between the plates is introduced to the software. To minimize mesh network near the water free surface in OpenFOAM software, two sheets of 0.1 m dimension meshes at 1.714 and -1.453 m elevations in z direction are applied. Accordingly, 1000,700 and 50 meshes are applied in x, y and z directions, respectively. To increase the simulation accuracy around the Chabahar bay, 200 extra meshes of 500 m size are applied at the bay mouth, located at 60 to 61 longitude and 25 to 25.5 latitude coordinate. The simulation results in terms of water surface undulation consequence are shown in Fig. 8, where ten extracted snapshot of the wave propagation is shown from the beginning, $t = 0$ s to $t = 3500$ s.

Note: the video file of tsunami propagation in Indian Ocean and Oman Sea is attached to the manuscript.

The tsunami forehead velocity profiles in a number of consequences during propagation is shown in Fig. 9, where the variation of the velocity impinge to the shore line and its depth variation are shown. The velocity escalates three times at half depth.

The estimating the time lags between the subduction occurrence and the arrival of the tsunami thereof at the coastline, and the run-up height of tsunami constitute the two vital parameters, necessary for tsunami early warning system emergency planning and for coastal zone management. The predicted maximum run-up height and arrival time of 1945 Makran tsunami at a number of major ports placed at the Indian Ocean and Oman Sea coastlines are tabulated in Table 2. The locations of the ports at the coastline are shown in Fig. 10.

According to Table 2, the tsunami reaches within 1000 s (less than 17 min) to Ormara port at Pakistan coastline, with 4.25 m run up height. There is no reliable data of 1945 Makran tsunami in order to verify the model through. According to Ambraseys and Melville (1982), the historical evidences reveal that tsunami wave run up reached up to 4 m in Pasni, which is consistent with the result of the model, which is 3.95 m, Table 2.

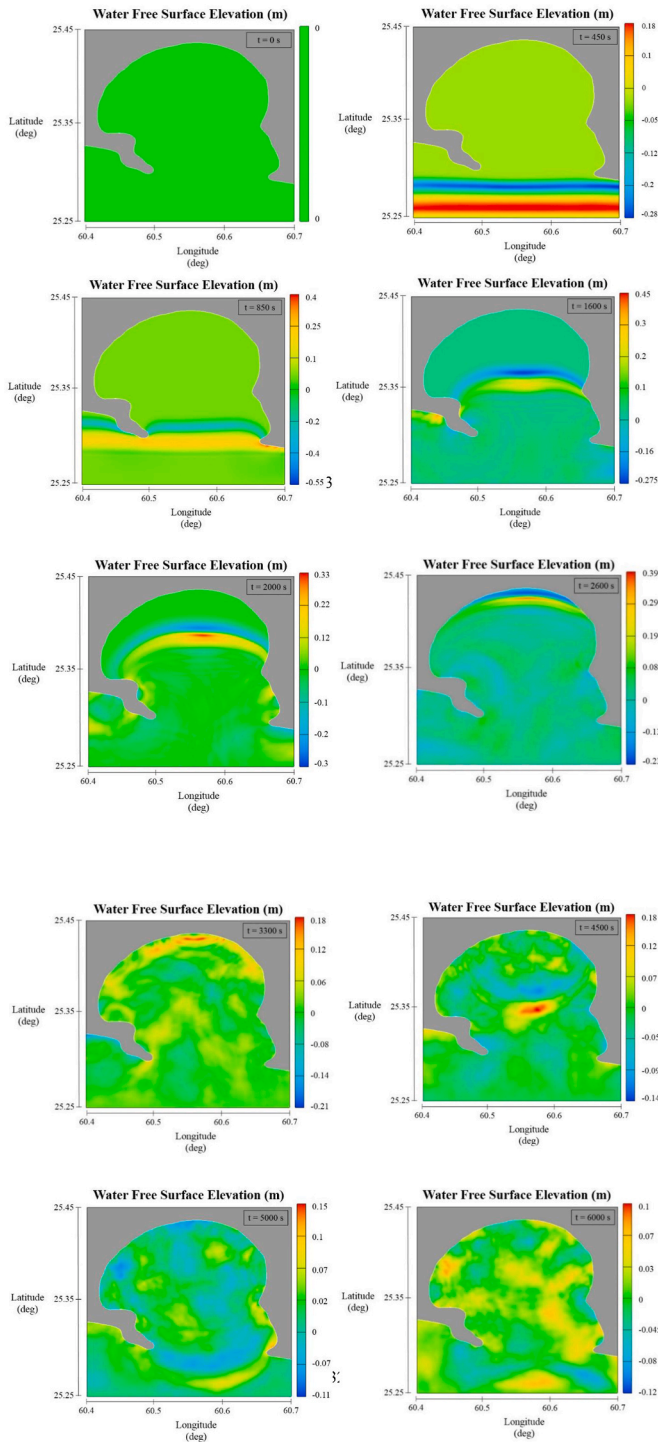


Fig. 13. Steps in tsunami wave propagation in the Chabahar bay, the results are extracted from the OpenFOAM regional model simulation.

4.2. Regional model

To estimate the pattern and characteristic features of the MSZ tsunami propagation in Chabahar bay in a more accurate manner, required for coastal management purposes, a regional model is run. The time series of free surface oscillation at the mouth of the bay, extracted from the global model is inserted as the input in the regional model. The wave train, (i.e. the time series of free surface oscillation (η)) due to the Makran subduction at the mouth of Chabahar bay at 22.7 m water depth, extracted from the global model is shown in Fig. 11. The time series consist of the main tsunami undulation, in the red box, and the series of successive reflective waves, the main part of which is considered for the regional modelling input.

To run the simulation process in OpenFOAM software in this stage, two sheets of 0.01 m mesh at 0.2 and -0.3 m elevations in z direction are applied. The 1200, 1200 and 150 meshes are applied in x , y and z directions, respectively. Same as the global model mesh independence studies are run to obtain an optimal mesh resolution. The result is shown in Fig. 12. The dimensions of optimal mesh size applied in the regional model is 25.53 m.

The simulation results in terms of water surface oscillation consequence are shown in Fig. 13, where ten extracted snapshot of the wave propagation are shown from the beginning, $t = 0$ (time is reset at Chabahar bay mouth) to $t = 6000$ sec.

Note: the video file of tsunami propagation in Chabahar bay is attached to the manuscript.

According to the regional model results, the maximum run-up at the port of Chabahar and the arrival time of the tsunami wave are 0.475 m and 2220 s, respectively. For the Chabahar port the estimated run up from the regional model is 15.8% higher than that of the global model, 0.398 m, Table 2, indicating that applying the consequent modelling precision improves the results, significantly.

4.3. Tsunami risk zone of Teess beach town

Prediction of maximum run-up height on different slopes and the maximum horizontal excursion is efficient for engineering applications. To implement the results of the 3D simulation in an engineering practice, this study is developed to determine the tsunami risk zone of the planned new beach town of Teess at the Northern coast of Chabahar, Fig. 14. The slope varies within 0.4%–1.5% range, from deep water to coastline and above, respectively, along section A-A, Fig. 14.

To predict the probable effects of 1945 Makran tsunami on Teess coasts, the 2D OpenFOAM model with very fine mesh and accurate bathymetry is applied. For this purpose, the free surface oscillation time series (η) at 4 m depth is extracted from the 3D regional model, Fig. 15 and inserted as the input in the 2D model. For the 2D numerical simulation, 2500 and 300 meshes are considered in x and z directions, respectively.

The beach cross section at Teess beach town and the result of the model in terms of the tsunami undulation is shown in Fig. 16, where the maximum wave run-up ($R = 0.72$ m) and the maximum horizontal incursion of flood ($x = 34.6$ m), are shown in the figure. The engineering practice on assessing Teess inundation area and the affected flooded zone is further extended to the possible tsunami scenarios in terms of the possible location of subduction zone on Makran fault line and the probable earthquake magnitude.

5. Conclusion

Historically, more than 75% of worldwide tsunami events are generated by earthquakes. The coastlines of Oman Sea at western Indian Ocean region are vulnerable to tsunami threats because they are located near active earthquake faults of Makran subduction zone (MSZ), where the time of arrival for a tsunami can be very short. In this study, an advanced 3D numerical simulation package is proposed for predicting

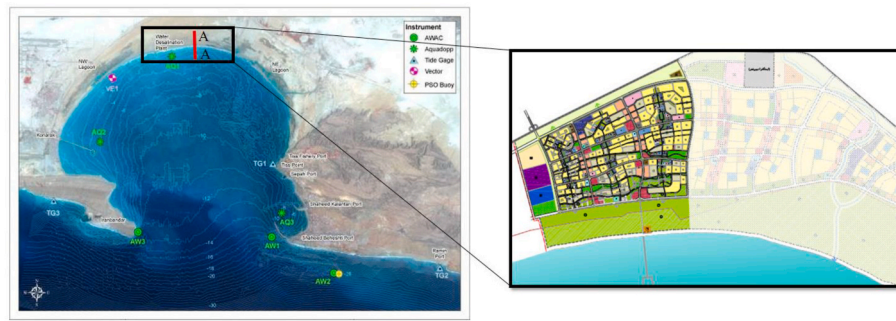


Fig. 14. Location and plan of the new beach town of Teess, phase 1 at Chabahar Bay.

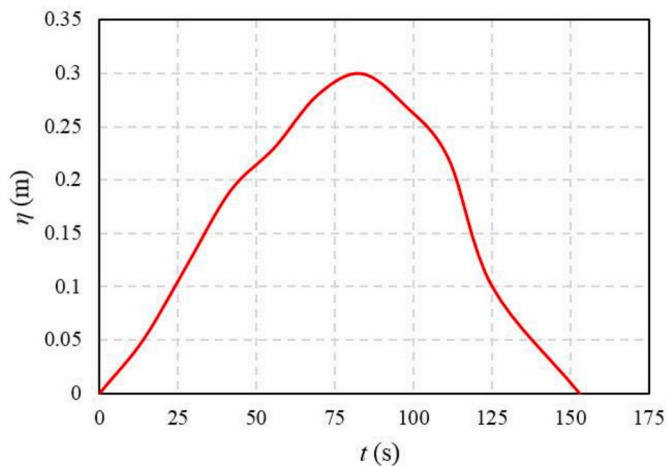


Fig. 15. The time series of the major free surface oscillation at 4 m depth of water in Chabahar Bay in front of Teess coastline.

generation and propagation of tsunami. The Okada algorithm for generation of original N-wave is combined with the Volume-Averaged Reynolds-Averaged Navier-Stokes solver, IHFOAM, developed in OpenFOAM s/w for 3D propagation of the wave. The approach is applied for 1945 Makran tsunami at Oman Sea, Indian Ocean. This 3D simulation provides more accurate results than the conventional 2D models based on the shallow water assumptions and Boussinesq-type

equations. The extensive comparison with laboratory data and limited available field data indicates that the model is promising in predicting the run up. To compromise between the accuracy and the time spent on 3D numerical modeling, the simulation is run in three stages, of global, regional and local, where coarse meshes of 1400 m–500 m size in global Indian Ocean and finer meshes of size of 25 m are applied in Chabahar Bay and very fine 2D mesh of 0.4 m is applied at and near the coastline. The total simulation run time is recorded through a desktop computer (cori7) is about 14 h, which is quite reasonable. The 16% difference of run up height at Chabahar port between the results of the regional and global models, indicates the sensitivity of the model to the bathymetry precision and the mesh size.

This 3D numerical package can be applied to the issues of tsunami propagation over different bottom geometries including the wave propagation over submerged obstacles. Because tsunamis originated from MSZ are the major threat for the Southern coasts of Iran, Pakistan, India and Northern coasts of Oman, further studies are required to assess the extent of hazard due to possible extreme scenarios (in terms of location of subduction on the fault line and magnitude of earthquake) and/or on the effectiveness of artificial depleting structures.

CRediT authorship contribution statement

Masih Honarmand: Simulation, Writing - original draft. **Ahmad Shanehsazzadeh:** Conceptualization, Methodology, Supervision, Writing - review & editing. **Sayed Mahdi Zandi:** Methodology, Supervision.

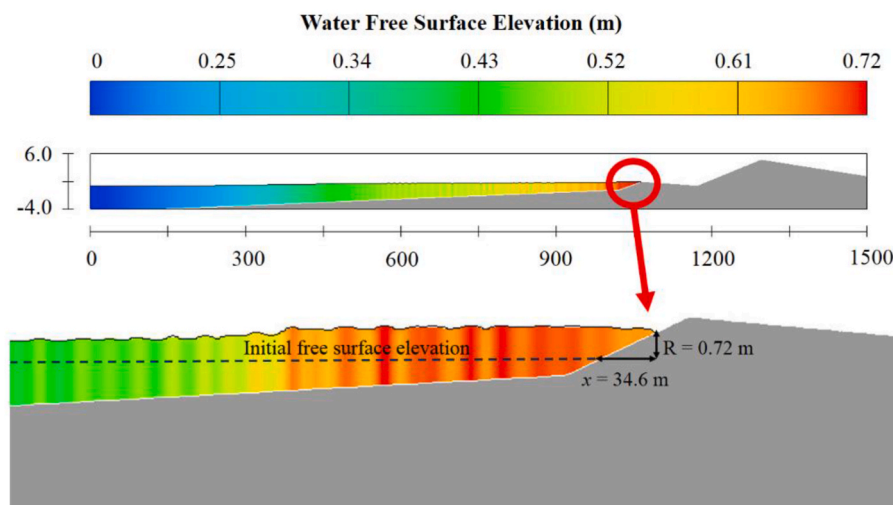


Fig. 16. Water surface rise at Teess beach town due to 1945 MSZ tsunami. The predicted maximum wave run-up (R) and the maximum horizontal excursion (x) are shown in the figure.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.oceaneng.2020.108109>.

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